

Attributing mortality among drivers of population decline in *Acropora palmata* in the Florida Keys (USA)

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Abstract *Acropora palmata* populations have experienced steep declines over the past 30 years. Although numerous culprits are recognized, their relative contributions to the decline are poorly quantified, making it difficult to prioritize effective conservation measures. In 2004, a demographic monitoring program was implemented in the Florida Keys (USA), aimed at determining the relative importance of various stressors affecting *A. palmata*. A subset of randomly selected *A. palmata* colonies within 15 fore-reef plots was tagged and surveyed three to four times per year over 7 years. Colony size, live tissue, prevalence of disease, snail (*Coralliophila abbreviata*) predation, physical damage and other conditions were assessed at each survey. The estimated effect of each condition causing recent mortality was ranked, and together, these parameters were used to attribute the population-level tissue loss associated with each condition. In addition, all new colonies in the study plots were counted and assessed annually in order to track trends in total colony count and live tissue abundance. Between 2004 and 2010, the study population has shown more than 50%

decline in live area from three main conditions: fragmentation, disease and snail predation. Approximately half of this decline occurred during the catastrophic 2005 hurricane season from which recovery has been minimal. Meanwhile, colony abundance has shown gradual decline throughout the study. Snail predation was the most prevalent condition. However, it ranked third in attributed tissue loss, behind breakage that occurred during the 2005 hurricane season, and disease. Thermal bleaching of *A. palmata* was not observed during this study. Because mortality continues to outpace recruitment and growth, intervention to ameliorate losses to the more manageable threats including predation and breakage could result in substantial conservation of live tissue, buying time for the abatement of less tractable threats to *A. palmata* recovery such as climate change and disease.

Keywords Bleaching · Coral disease · Damselfish · Corallivory · Hurricane damage · *Coralliophila*

Introduction

Population decline of acroporid corals has been documented throughout the Caribbean basin over the past three decades (Porter and Meier 1992; Garzon-Ferreira et al. 1993; Bruckner 2002a; Miller et al. 2002a, b; Bruckner and Bruckner 2003; Sutherland and Ritchie 2004). A general pattern of stasis has been punctuated by catastrophic events (Gardner et al. 2003) in the form of physical disturbances and bleaching/disease episodes (Goreau 1992; Aronson and Precht 2001). *Acropora palmata* is one of the dominant framework builders on western Atlantic reefs (Shinn 1989; Goreau 1992), forming the major geological structures of Florida Keys high-relief reefs (Shinn 1989). In terms of

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structural complexity and growth rates, acroporid corals are considered ecologically irreplaceable on western Atlantic reefs (Bruckner 2002a), lending urgency to the need for effective management/conservation to maintain ecologically functional populations. In 2006, *A. palmata* was listed as a threatened species under the US Endangered Species Act (NMFS 2006), providing legal mandate for ‘recovering’ this species.

Along with thermal mass bleaching events, the main causes for *Acropora* spp. decline are typically categorized as disease, fragmentation and predation (Bruckner 2002a; NMFS 2006). These latter three conditions are routine in all natural populations but occasionally escalate to catastrophic levels or can result in compensatory effects when populations are diminished (Knowlton et al. 1990; Jayewardene et al. 2009). Thirty years ago, white-band disease (WBD) began to spread throughout the Caribbean basin (Gladfelter 1982), and more recently, white pox (WPx) has been described and noted to have substantial regional impact on *A. palmata* populations (Rodriguez-Martinez et al. 2001; Patterson et al. 2002). However, additional acute tissue loss condition(s) not fitting the characteristic descriptions of either WPx or WBD are also reported (rapid tissue loss, RTL; Williams and Miller 2005; Williams et al. 2006, 2008). Fragmentation is a vital part of *Acropora* spp. lifecycles, allowing rapid proliferation to form mono-specific thickets in favorable settings. However, as abundance and live tissue cover declines, fewer fragments are retained on the reef (Baums et al. 2003; Williams et al. 2008) and have adequate tissue cover to successfully fuse to the substrate.

The corallivorous snail, *Coralliophila abbreviata* (hereafter referred to as ‘snails’), is the main predator of *A. palmata*, consuming up to $16\text{ cm}^2\text{ day}^{-1}$ (Brawley and Adey 1982) of tissue, leaving behind notable feeding scars on the colony (Miller 2001). Although *Stegastes planifrons* (hereafter referred to as ‘damselfish’) do not directly prey on *Acropora* spp., they remove live tissue in order to clear space for their territories (Kaufman 1977; Bruckner 2002a). Both of these ‘occupant species’ cause persistent tissue loss and potentially reduced fitness due to chronic costs of regeneration/repair (Henry and Hart 2005).

Resource managers require knowledge of population status, trends and interventions that will foster recovery for this imperiled species, but because *A. palmata* forms easily fragmenting, structurally complex colonies, it is difficult to define abundance. Typically, most reef-monitoring programs track only percent cover based on a 2D projected area. Unlike boulder corals, however, much of the live area of *A. palmata* is on the underside of branches, not visible in planar view. The morphological complexity of *A. palmata* makes direct quantification of changes in tissue area impractical (Bythell et al. 2001) on a long-term monitoring scale. Colony abundance is a useful but potentially

deceptive parameter for highly fragmenting organisms that frequently undergo physiological fission and fusion (Edmunds and Elahi 2007). Dramatic losses of live area are possible with no change, or even an increase, in colony abundance if partial mortality results in isolates which are then counted as separate colonies on subsequent surveys. Further, these isolates are often confused as sexual recruits (Miller et al. 2007), signaling a booming population instead of a dying one.

The high temporal and spatial variability in the prevalence and severity of various threats make it difficult to discern their relative importance to the population over time. Based on our demographic approach, we can track individually defined colonies and calculate the prevalence (% of individuals affected) of various threats at greater frequency than most long-term coral monitoring programs. However, the prevalence alone does not reveal the true impact of the condition on the population. It is unclear, for example, if the chronic effects of predation could outweigh intermittent acute disease events and how these compare to the catastrophic losses associated with hurricanes. An understanding of the relative magnitude of these effects is important for prioritizing management and conservation efforts (Bruckner 2002b). Because attributing relative amounts of tissue area loss for large colonies afflicted with multiple sources of mortality has not been attempted in other long-term monitoring efforts, somewhat novel methods are required.

This study describes a 7-year implementation of a demographic monitoring protocol in the upper Florida Keys developed to address many key population metrics as well as trends in abundance and pervasiveness of various threats in remnant populations of *A. palmata* (Williams et al. 2006). To better characterize population abundance, we employ explicit criteria for consistent ‘colony’ delineation, along with tractable field visual estimates yielding an index of tissue area. Additionally, we used a ranking system to partition tissue loss attributable to the various sources of mortality. Thus, here, we examine (1) changes in the abundance of both colonies and live tissue; (2) prevalence of common conditions affecting *A. palmata*; and (3) the relative importance of these conditions in driving population changes.

Materials and methods

Survey methods

In 2004, five fore-reef sites were selected in the upper Florida Keys (USA). A total of 15 fixed circular study plots (150 m^2) were established (Table 1) on reef spurs with at least 10 distinct *A. palmata* colonies. Colonies were initially delineated based on continuous live tissue or

Table 1 Study plot locations and general characteristics

Reef	Plot	Latitude (°N)	Longitude (°W)	Depth (m)	# Tagged colonies		Total # colonies		Avg dimension (cm)		Avg colony % live	
					Apr-04	Oct-10	Oct-04	May-10	Oct-04	May-10	Oct-04 (%)	May-10 (%)
Carysfort	CF1	25°13'19.0"	80°12'38.0"	3	14	3	17	7	59.1	37.1	64	75
	CF2	25°13'18.4"	80°12'38.2"	3	12	9	37	24	53.3	65.6	71	44
	CF3	25°13'22.4"	80°12'34.4"	4	12	4	11	8	93.5	97.0	68	53
Elbow	EL1	25°08'33.3"	80°15'30.1"	4	13	8	43	60	56.4	39.1	77	74
	EL2	25°08'34.4"	80°15'29.6"	4	12	9	26	27	62.7	42.8	62	51
	EL3	25°08'38.2"	80°15'28.1"	5	12	7	31	17	45.4	47.2	72	57
	EL4	25°08'42.3"	80°15'26.4"	6	17	6	21	10	39.3	63.1	82	76
	EL5	25°08'42.7"	80°15'26.6"	6	12	3	13	5	32.5	43.3	90	53
French	FR1	25°02'02.1"	80°20'57.9"	7	12	10	28	38	68.3	56.0	82	41
Key Largo Dry Rocks	KL1	25°07'25.0"	80°17'50.5"	3	13	8	17	8	81.7	84.2	58	22
	KL2	25°07'22.4"	80°17'52.3"	2	12	0	27	0	40.6	–	63	–
	KL3	25°07'21.2"	80°17'53.7"	4	12	7	15	13	43.5	36.3	54	31
Molasses	ML1	25°00'34.5"	80°22'29.3"	4	10	7	11	9	66.6	51.6	72	67
	ML2	25°00'32.8"	80°22'29.0"	6	18	8	23	24	48.6	39.8	65	66
	ML3	25°00'36.5"	80°22'23.8"	4	11	10	22	21	56.0	60.5	94	63
					$\Sigma = 192$	$\Sigma = 99$	$\Sigma = 342$	$\Sigma = 271$	$\bar{X} = 56.5$	$\bar{X} = 54.5$	$\bar{X} = 72$	$\bar{X} = 55$

A random subset of colonies was tagged for assessment 3–4 times per year beginning in April 2004, and all colonies in each plot were mapped beginning in October 2004 and annually in spring thereafter. Average dimensions and % live are shown for all colonies found (both tagged and untagged) within the 150 m² study plot. Note that an additional 18 colonies were tagged among the 15 study plots in 2006 and are included in the total of 99 shown for October 2010. Only 94 of the original tagged colonies remained in October 2010

discontinuous patches of tissue that were occupying the same skeletal structure (detailed in Williams et al. 2006, 2008). For plots with greater than 18 colonies, a subset of 12 were randomly chosen and tagged; an additional 18 ‘replacement’ colonies were tagged in fall 2005 among the plots where some had died. Thus, the placement of the plots was not random (i.e., targeting where colonies occurred), but the selection of colonies within plots was random. The subset of tagged colonies was surveyed three to four times per year from April 2004 to Oct 2010. In addition, all colonies within the study plots were mapped and photographed once per year to re-identify colonies present the previous year and any new colonies, including asexual or sexual recruits that were attached to the substrate.

Annual surveys entailed counting and measuring in situ the length (L, maximum diameter), width (W, longest axis

perpendicular to length) and height (H) of the skeletal structure of all colonies, including all new recruits, to the nearest centimeter, and visually estimating the percent of live area of the measured colony skeleton (% live) to the nearest 5%. Fragments were only included as colonies after they self-attached to the substrate. At each interim survey, only the subset of tagged colonies was measured and visually assessed for several parameters including bleaching (the presence of any pale or white areas where tissue was intact) and the presence and abundance of the occupant species, *Coralliophila abbreviata* (corallivorous snail) and adult *Stegastes planifrons* (threespot damselfish). The severity of and conditions associated with recent mortality (RM) were also assessed for each tagged colony. Recent mortality (RM) was distinguished as exposed white skeleton with little or no algal colonization and sharply defined corallites (with the exception of parrotfish bites).

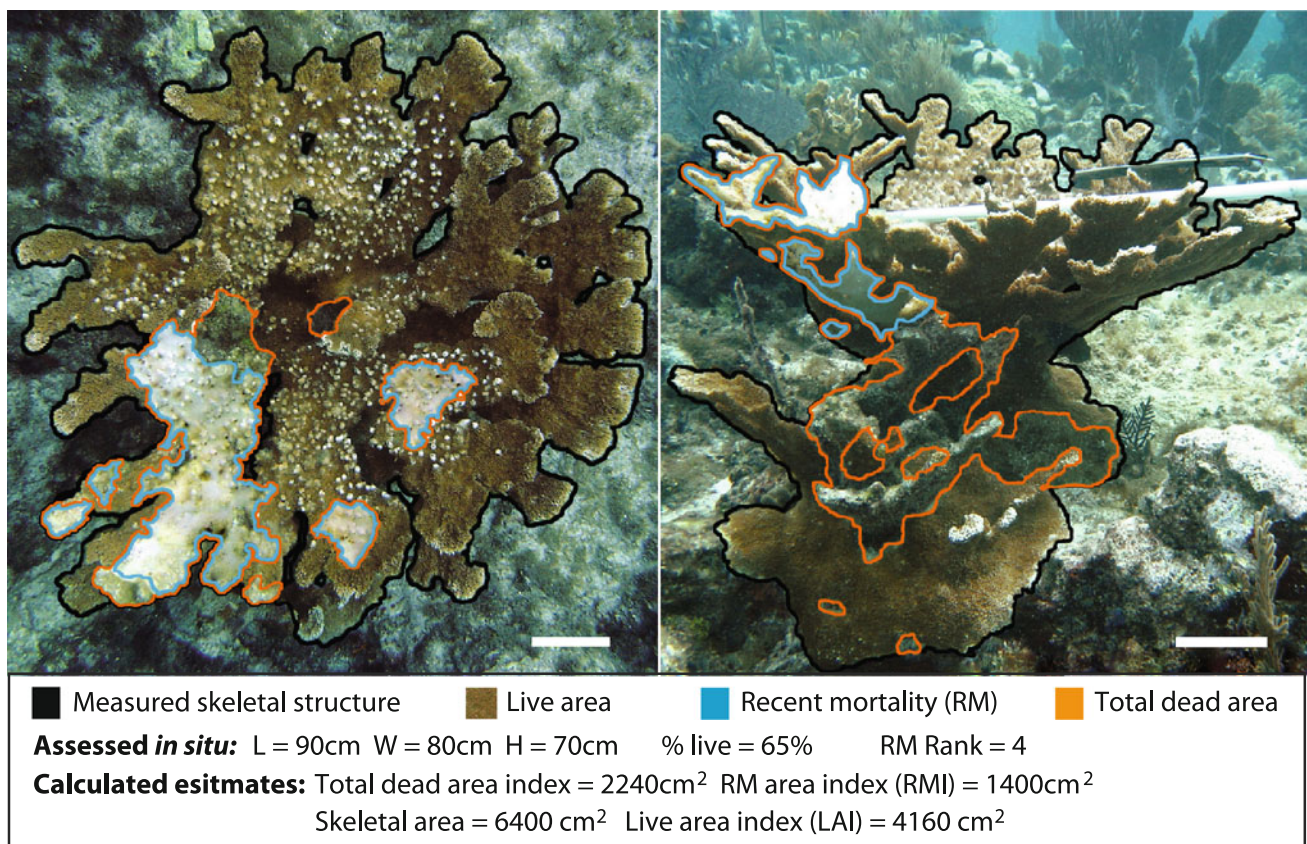


Fig. 1 Schematic illustrating *A. palmata* colony measurements and calculations. The length (L), width (W) and height (H) of the delineated colony skeleton were measured, and the % live of the complex skeletal structure (outlined in black) was visually estimated in situ (outlined in photograph only for illustration, not analysis). Severity of recent mortality (RM) was visually ranked based on the ratio of recent (blue) to total (orange) dead area. In this illustration, the colony skeleton has an average dimension of 80 cm (scale bar in photo is 10 cm) and was estimated to be 65% live (35% dead). The RM (blue) was ranked as ‘4’ or 50–75% of the total dead area

(orange). The live area index (LAI) is calculated as the average dimension squared (80 cm × 80 cm = 6,400 cm²) and then adjusted for the % live (65%) resulting in an LAI of 4,160 cm² leaving 2,240 cm² of the skeletal area as the total dead area index. For RM rank 4, the midpoint between 50 and 75% (62.5%; ESM, Appendix 1) is used to estimate the RM Area Index (i.e., 62.5% of the total dead area [2,240 cm²] or 1,400 cm² as the RMI). All 1,400 cm² of the RMI was attributed to white disease, the only condition observed on this colony

Table 2 Explanation of observed and calculated entities and indices as defined for this study

Entity	Definition
Observed in situ	
Colony (L,W,H)	The delineation of a three-dimensional skeletal feature defined as an “individual” measured as length (L), width (W), height (H)
% Live/dead	Visually estimated percentage of the measured colony surface that is covered/lacking live tissue
Parameter	Characteristics of colonies which are not associated with RM including prevalence of bleaching, and the prevalence and abundance of occupant species (<i>Coralliophila abbreviata</i> and <i>Stegastes planifrons</i>)
Recent mortality (RM)	Delineated as presence/absence of portions of the colony lacking live tissue but still having sharply defined corallite structure and minimal algal colonization
RM rank	When RM is present, a rank based on the proportion of the dead area that is RM (Rank 1 $\geq$ 0–5%; 2 = 10–20%; 3 = 25–45%; 4 = 50–75%; 5 = 80–100%)
Condition(s)	Directly observable sources of recent mortality including white disease, snail feeding scars, ciliate infestation, <i>Cliona</i> lesions, parrotfish bites and fish fecal matter
Calculated metrics/indices	
Average dimension	$(L + W + H)/3$
Skeletal area	Calculated as the average dimension squared
Live area index (LAI)	Calculated as the skeletal area multiplied by % live to account for only the live portions of the colony
Total dead area index	Calculated as the skeletal area multiplied by the % dead (as a proportion) to reflect only the dead portions of the colony. The % dead is simply the converse of the %live (100-%live)
Recent mortality area index (RMI)	Total Dead Area index multiplied by the midpoint percent of the range represented by the RM rank (See Appendix). This represents negative change in LAI or “tissue loss”. This index is summed among colonies to estimate ‘tissue loss’ attributable to each condition
Skeletal breakage	Negative change in skeletal area between two surveys. Calculated as negative change in area multiplied by the % live at the previous survey based on the assumption that the broken portion had the same percent live as the previously intact colony

See Fig. 1 for graphic illustration

The severity of RM was ranked on a scale of one to five (Electronic Supplemental Material, ESM, Appendix 1) corresponding to the visually estimated proportion of the dead area that was recently dead (Rank 1 for >0 –5%, Rank 2 for 10–25%, Rank 3 for 30–45%, Rank 4 for 50–75%, and Rank 5 for 80–100%). For example, given a colony is estimated as 65% live, if the dead portion (35%) consists of between one-half and three-quarters RM, then this colony would be assigned a rank of 4 in the field assessment. If the entire dead area was RM, then it would be ranked a 5. Hence, the ranking was assessed as the ratio of RM area to total dead area on the colony (see Fig. 1).

The apparent source(s) of the RM was categorized as one of the following conditions: white disease, snail feeding scars, ciliate infestation (Croquer et al. 2006), lesions associated with the boring sponge, *Cliona laticavicola*, biting by parrotfish (typically *Sparisoma viride*) and lesions resulting from fish fecal matter smothering the tissue. Three types of ‘white disease’ were distinguished in this study as follows: white pox (WPx) disease, white-band disease (WBD) and rapid tissue loss (RTL). However, because the distinction between these diseases is ambiguous, they were pooled as ‘white disease’ for the purposes of analyses. Additional rare conditions causing RM included

direct competitive overgrowth (e.g., by *Halimeda* spp. or *Erythropodium caribaeorum*), *Hermodice carunculata* feeding scars, bleaching, abrasions and damselfish activity (lawns and nests). Unidentified sources were categorized as ‘other’ in the analyses. All conditions were identified visually based on gross characteristics with further descriptions and illustrations found in Williams et al. (2006). In cases where more than one condition was observed causing recent mortality, they were recorded as primary, secondary and tertiary based on the relative amounts of recent mortality associated with each.

Data analysis

Total colony and tissue abundance

Three population metrics were calculated for each of the 15 study plots at each annual survey: (1) the total number of colonies in the study plot, including newly attached fragments or sexual recruits, (2) the average colony dimension $[(L + W + H)/3]$ of all attached colonies in the study plot and (3) the sum of live area index (LAI; Table 2) for all attached colonies in the study plot. The LAI was calculated as the average colony dimension squared (an index of skeletal area) multiplied by the proportion of live tissue

cover to adjust for partial mortality, i.e., $((L + W + H)/3)^2 \times \% \text{ live}$) yielding an index corresponding to live tissue area (m^2 LAI; see Fig. 1 for example). Linear regression was used to evaluate the impact of the 2005 hurricane season and to identify post-hurricane recovery (or lack thereof) observed in these three metrics. The three surveys from fall 2004 to spring 2006 (or ‘storm interval’) were regressed to evaluate storm impacts and the five surveys beginning with spring 2006 through spring 2010 (‘post-storm interval’) were used in a separate regression to evaluate recovery trends (ESM, Appendix 2).

Tissue loss attribution

To evaluate the relative impact of the various observed sources of recent mortality (i.e., conditions), the RM rank assigned to each tagged colony during field surveys was converted to an estimate of the area of RM tissue loss (i.e., RM Area Index or RMI) as follows (Fig. 1, Table 2). Similar to the LAI calculation, an index of total dead area was calculated using the skeletal area and the converse of the % live estimate (i.e., 100-% live). This total dead area was then multiplied by the midpoint percentage associated with the RM rank (i.e., Rank 1 = 2.5%, Rank 2 = 15%, Rank 3 = 35%, Rank 4 = 62.5% and Rank 5 = 90%; see ESM, Appendix 1), yielding an index of the area of recent mortality (RMI; Fig. 1). RMI area was then attributed to the associated condition(s) observed in the field. In cases where two conditions were observed, two-thirds (66%) of the calculated RMI area was allocated to the primary condition and one-third (33%) to the secondary condition. When three conditions were observed, five-eighths (62.5%) was allocated to the primary condition, one-quarter (25%) to the secondary and one-eighth (12.5%) to the tertiary (see ESM, Appendix 1). The RMI area attributable to each condition for each tagged colony was summed over all sites for each survey to compare the total live area lost among conditions.

To determine the tissue loss associated with skeletal breakage, the difference between skeletal areas at each survey was calculated for each colony. In cases where the skeletal area decreased, the live versus dead tissue cover of lost skeleton was assumed to be proportional to the % live estimated for the colony at the previous survey, and the skeletal area loss was adjusted accordingly to estimate the tissue area lost to breakage (Table 2). This ‘tissue loss’ due to fragmenting of the tagged colonies might not truly be lost to the population if the fragments survive and grow. This addition to the population would be captured in the total plot colony and tissue abundance measurements assessed annually.

Spatial and temporal patterns

Annual, seasonal and spatial variation in the prevalence of bleaching and sources of recent mortality, as well as both prevalence and abundance of occupant species, was analyzed using the Friedman test for repeated measures with IBM SPSS Statistics 19 software. For the seasonal analyses, surveys conducted from March to May were categorized as spring, June to August were considered summer, September to November were considered fall, and December to February were considered winter. Because surveys of the 15 study plots spanned on average 21 days, the median date of the surveys was used to determine the season. Data were pooled to achieve a balanced design; for annual comparisons, data for all seasons were pooled ($n = 15$ study plots); for seasonal comparisons, data for years 2004–2009 were pooled ($n = 15$ study plots); and for site comparisons, all plots were pooled ($n = 19$ surveys). For all Friedman tests, significant differences ($\alpha = 0.05$) were followed by pairwise comparisons for all possible pairs (paired t -tests on mean ranks) using IBM SPSS Statistics 19 software. Bonferroni adjustment was used to evaluate significance in the pairwise contrasts between sites ($P < 0.005$), years ($P < 0.003$) and seasons ($P < 0.008$).

Results

Total colony and tissue abundance

The change in number and average dimension of all colonies is summarized by plot in Table 1. We observed an overall loss of 53% of the 210 tagged colonies between 2004 and 2010; 32% were found dead (at least part of the original skeleton still attached to the substrate but no live tissue), and 21% were entirely dislodged from the substrate. There was no significant trend in total number of colonies (Table 1, Fig. 2; ESM, Appendix 2a), during either the storm interval or the post-storm interval. The average dimension of all live attached colonies in the study plots was 56 cm (± 4 cm SE, $n = 15$) at the start, decreased significantly to 43 cm (± 4 cm SE, $n = 15$; Regression, $P = 0.025$, $r^2 = 0.1108$, slope = -0.0214) during the storm interval, but showed no significant increase during the post-storm interval (ESM, Appendix 2b). Similarly, LAI declined significantly, by more than 50% (Fig. 2; ESM, Appendix 2c), during the storm interval (Regression, $P = 0.007$, $r^2 = 0.1539$, slope = -0.0065) but showed no significant trend during the post-storm interval.

Fig. 2 Average ($\pm$ SE, $n = 14$ –15 plots) *A. palmata* colony diameter and abundance (right axis) and plot live area index (left axis). Study period was divided into a storm interval and post-storm interval for regression analysis (ESM, Appendix 2). *Single asterisk*-Hurricane Dennis, *double asterisk*- Hurricanes Katrina, Rita and Wilma

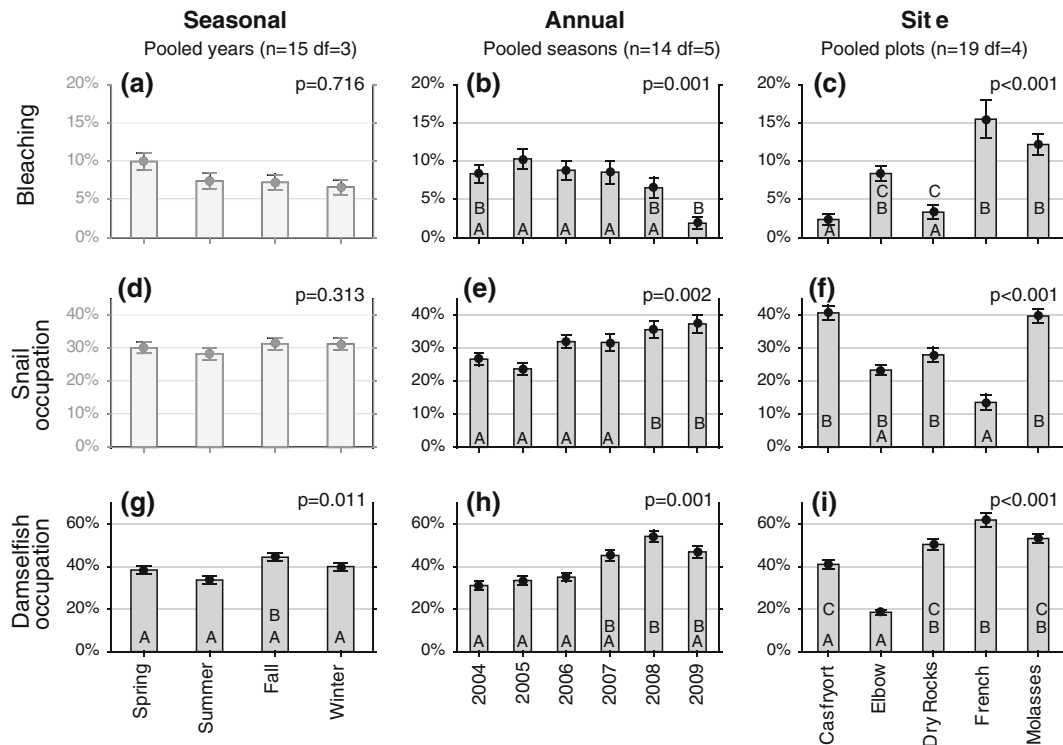
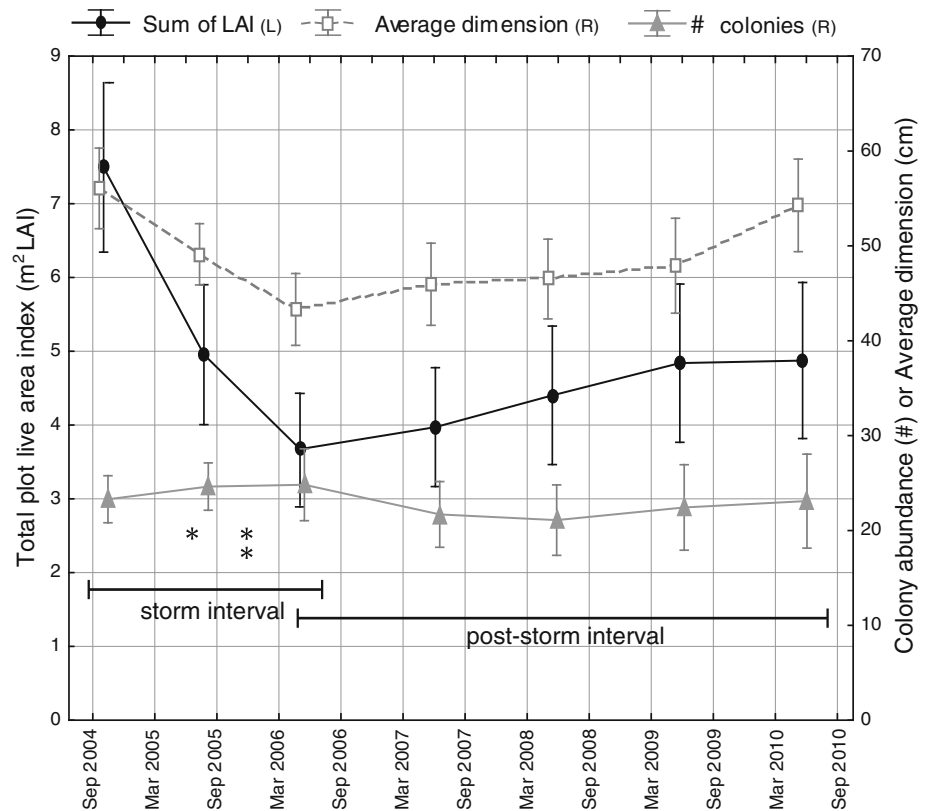


Fig. 3 Seasonal, annual and site variation in prevalence (mean $\pm$ SE) of colony parameters, including bleaching (a–c) and colony occupation by *Coralliophila abbreviata* (d–f; snail) or *Stegastes planifrons* (g–i; damselfish). Friedman tests on ranked data performed for each factor (P values indicated on each graph; sample size (n) and degrees of

freedom (df) indicated at the top of the column). Grayed out graphs reflect non-significant ($P > 0.05$) Friedman tests. Bars with the same uppercase letter did not differ significantly ($P > 0.008$ for season pairs, $P > 0.003$ for year pairs, $P > 0.005$ for site pairs) in post hoc pairwise comparisons

Fig. 4 Abundance and density of *Coralliophila abbreviata* snails on the population of tagged *A. palmata* colonies. Abundance is expressed as the sum of the number of snails on all the tagged colonies (averaged by plot, $\pm$ SE), and density is standardized to the summed live area index (LAI) of the tagged colonies in each plot (average $\pm$ SE)

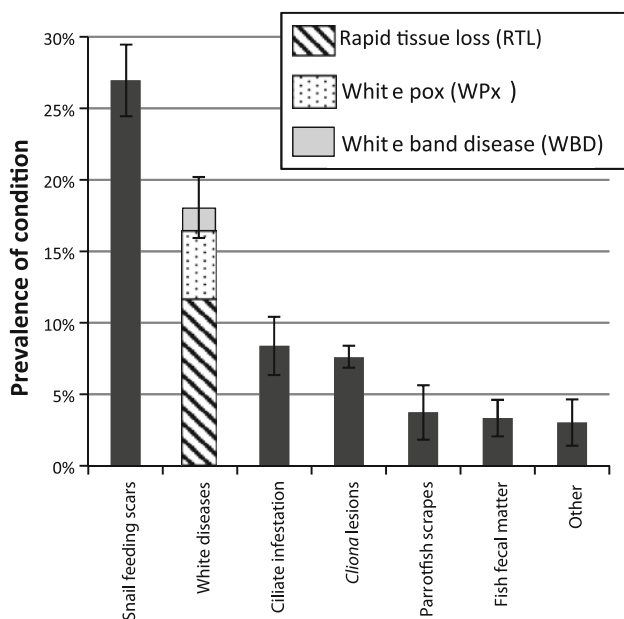
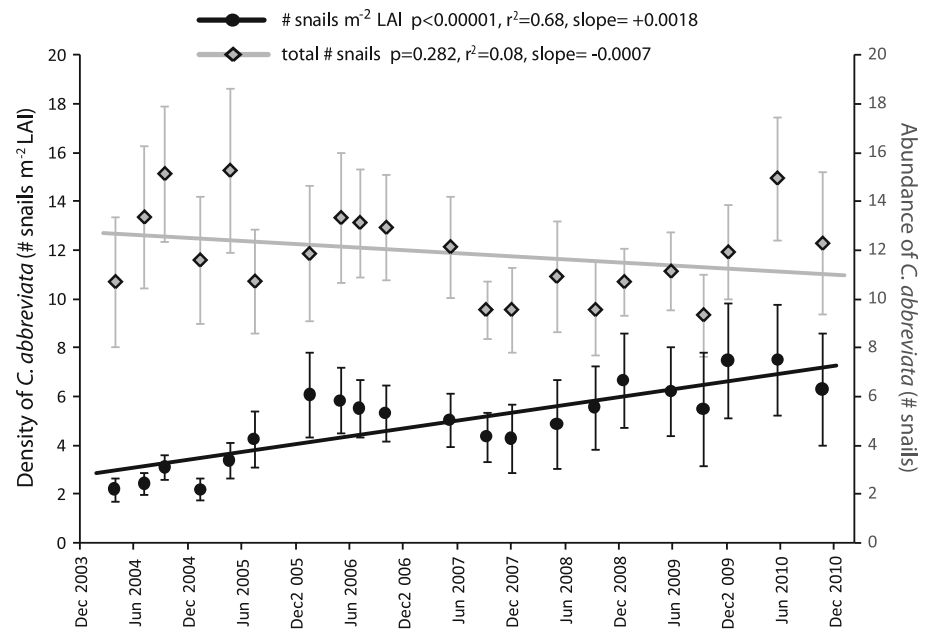


Fig. 5 Prevalence (% of tagged colonies affected, mean $\pm$ SE) of various recent mortality conditions affecting *A. palmata*. ‘Other’ category includes mortality associated with bleaching, overgrowth, abrasions, damselfish activity and unidentified conditions. Snail feeding indicates predation by *Coralliophila abbreviata*

Patterns in colony parameters

Bleaching

No mass bleaching events were observed during this study (Fig. 3a–c); in most cases, the observed bleaching affected only small patches of pale or bleached tissue and typically

affected fewer than 10% of colonies. Bleaching prevalence varied significantly between sites and years but not among seasons (Fig. 3a–c). Although there was no significant seasonal pattern to the mild bleaching that was observed, bleaching was routinely observed in one study plot at French Reef in the spring, when water temperatures were near their annual lows.

Prevalence and abundance of occupant species

Coralliophila abbreviata snails occupied 31% ($\pm$ 11% SD) of tagged colonies averaged over all surveys (2004–2010) and study plots, with on average 4.1 ($\pm$ 3.5 SD, $n = 875$) snails per occupied colony (median = 3.8), pooled across surveys. Snail occupation (% of colonies with at least one snail; Fig. 3d–f) varied significantly between years (Fig. 3e). Significant increases in the number of snails per occupied colony could not be discerned due to the high variability. However, when the number of snails is standardized per m^2 of host LAI, there has been a significant increase (Regression, $P < 0.0001$, $r^2 = 0.68$, slope = +0.0018) over the study while the total number of snails on all tagged colonies did not change significantly (Fig. 4).

On average, *Stegastes planifrons* damselfish were observed on 37% ($\pm$ 24% SD) of tagged colonies with on average 1.6 ($\pm$ 1.2 SD, $n = 1,116$) fish per occupied colony (median = 1.3). The prevalence of damselfish-occupied colonies was significantly higher in 2007–2009 compared with that in 2004–2006 (Fig. 3h). The more northern sites (Elbow and Carysfort) had significantly lower prevalence of damselfish-occupied colonies than the more southern sites (Fig. 3i).

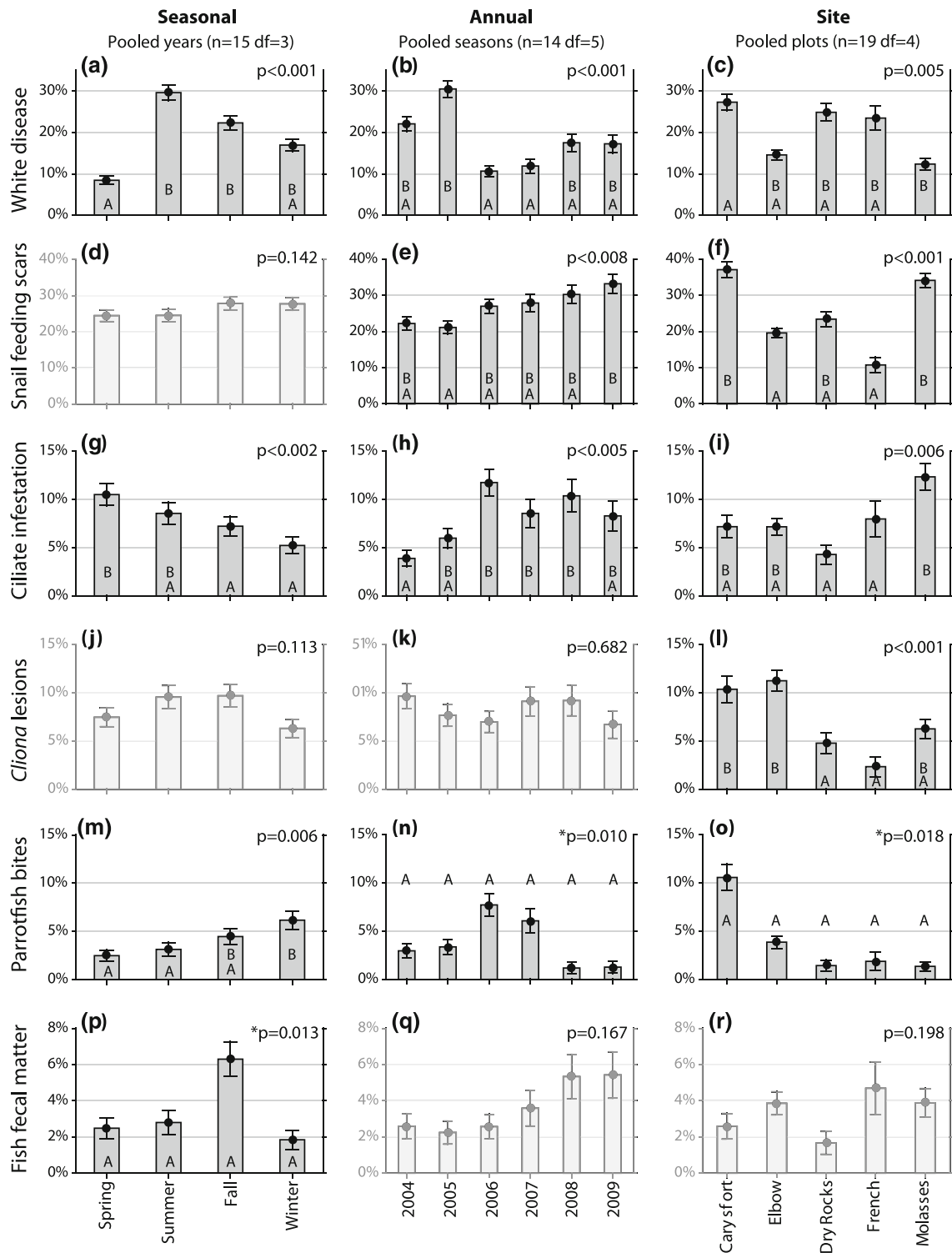
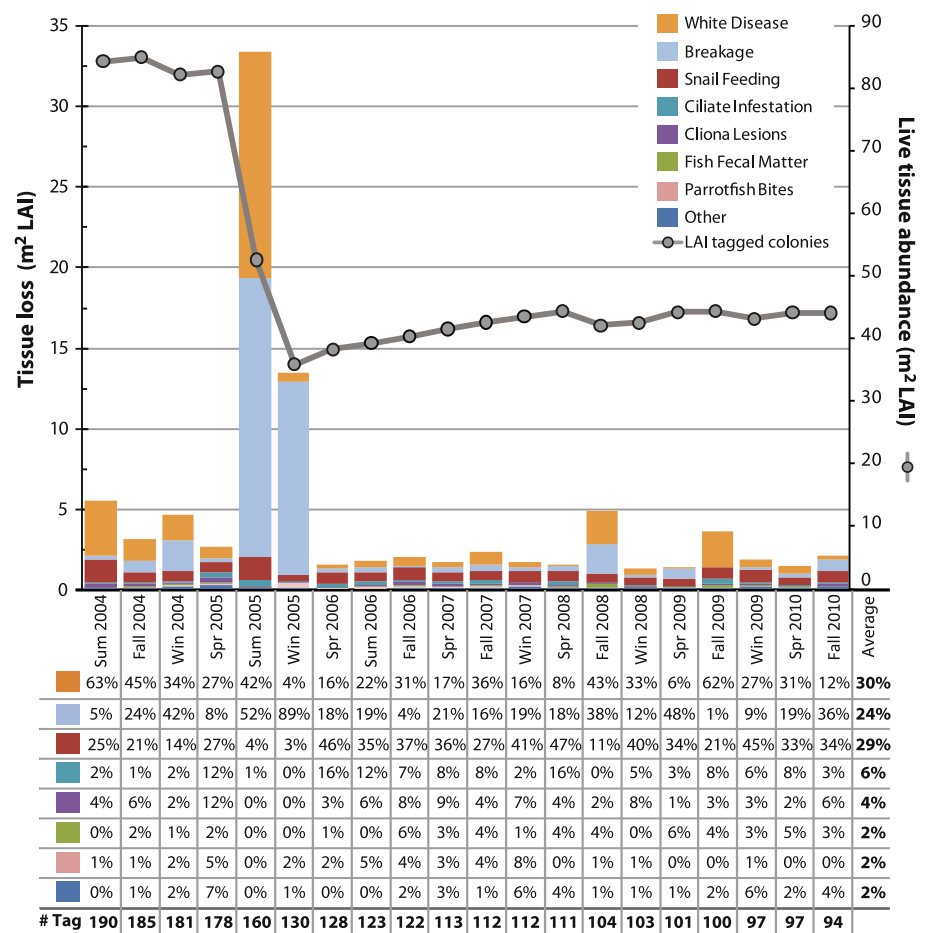


Fig. 6 Seasonal, annual and site variation in the prevalence (mean $\pm$ SE) of conditions associated with recent mortality. Friedman tests on ranked data performed for each factor (P values indicated on each graph; sample size (n) and degrees of freedom (df) indicated at the top of the column). Grayed out graphs reflect non-

significant ($P > 0.05$) Friedman tests. Bars with the same uppercase letter did not differ significantly ($P > 0.008$ for season pairs, $P > 0.003$ for year pairs, $P > 0.005$ for site pairs) in post hoc pairwise comparisons. Asterisk, no significant pairwise contrasts

Fig. 7 Total tissue loss and live area index (LAI) for the subset of tagged *A. palmata* colonies at each survey summed across plots. The total for each survey is partitioned by breakage and condition (shown as a percentage in the table). The number of tagged colonies alive at each survey is shown on the bottom row



Patterns in tissue loss conditions

When averaged over all sites and survey periods, snail feeding was the condition most commonly associated with recent mortality, followed by white disease, ciliate infestation and lesions associated with the excavating sponge *Cliona laticavicola* (Fig. 5). Recent mortality associated with parrotfish scraping and from fish fecal matter was observed on fewer than 5% of colonies on average. Of the three white diseases, RTL was the more prevalent followed by WPx and WBD (Fig. 5). Although each of the RM conditions varied on different spatial and temporal scales (Fig. 6), none of the sites had consistently lower prevalence of all conditions and parameters (Figs. 3, 6, right columns). Significant seasonal differences (Fig. 6, left column) were found in the prevalence of white disease, ciliate infestation and parrotfish bites. Ciliate infestation was significantly greater in the spring and summer (Fig. 6g), white disease prevalence was significantly greater in summer than in spring (Fig. 6a), and parrotfish scraping was significantly greater in fall and winter surveys than in the rest of the year (Fig. 6m). White disease prevalence was significantly greater in 2004 and 2005 than in

2006 and 2007, while 2008 and 2009 were intermediate (Fig. 6b).

Tissue loss attribution

The tissue loss associated with RM conditions and breakage of the subset of tagged colonies totaled 90-m² LAI over this 7-year study. Skeletal breakage accounted for an estimated 42% (38 m² LAI) of the total loss, white disease accounted for 33% (30 m² LAI) and snail feeding accounted for 15% (14 m² LAI). On average across surveys, 30% of the tissue loss at each interval was attributed to white disease, 24% to breakage and 29% to snail feeding (Fig. 7). Lesions associated with *Cliona* sp. and ciliate infestation resulted in approximately 5% of the total tissue loss on average at each survey period, and the remaining conditions were estimated to cause less than 3% of the loss (Fig. 7).

Of the total estimated 90-m² LAI tissue loss, more than half (47 m² LAI) occurred during the 2005 hurricane season. When this catastrophic period (summer and winter 2005 surveys, see Figs. 2, 7) is excluded, white disease accounted for 36% (16 m² LAI), breakage accounted for

20% (9 m² LAI) and snail feeding accounted for 27% (12 m² LAI) of the total tissue loss. Over the 20 survey intervals, these three conditions have ranked as the top three in all but four surveys (Fig. 7).

In addition to the catastrophic breakage associated with the 2005 hurricane season, notably elevated breakage was detected during the winter 2004 and fall 2008 surveys (Fig. 7). Tissue loss associated with white disease ranged from 0.1 to 14.0 m² LAI at individual surveys. Aside from the extreme tissue loss event that accompanied Hurricane Dennis in 2005 (Williams et al. 2008; Fig. 7), it is possible to discern some episodes of relatively greater disease-attributed tissue loss (e.g., summer 2004, fall 2008 and fall 2009) superimposed on a lower but persistent background level of tissue loss ranging from 0.1 to 1.5 m² LAI (Fig. 7). Tissue loss associated with snail feeding was relatively constant with no discernable ‘episodes’, and the loss observed at each survey ranged from 0.4 to 1.4 m² LAI per survey interval, which was similar to the background range for white disease.

Discussion

Since 2004, the live area of the *A. palmata* population in the upper Florida Keys has declined by approximately 50% (Fig. 2), mainly from the following three conditions: breakage, disease and snail predation. The four hurricanes affecting the Florida Keys in 2005 caused widespread breakage, and the first storm, Hurricane Dennis, was followed by dramatic white disease-associated tissue loss (Fig. 7; Williams et al. 2008). Similar storm-associated declines of approximately 50% in *Acropora* cover (Stoddart 1974; Rogers et al. 1982; Rogers 1993; Lirman and Fong 1996) have been observed in other studies. Although acroporid life history is well adapted to disturbance, relatively little asexual recruitment was observed in spring 2006 (Williams et al. 2008), and colony abundance did not change significantly over the course of the study (Fig. 2). The bulk of the loss of live area can be explained by partial colony mortality as indicated by the decrease in the average % live (Table 1), emphasizing that colony abundance alone is a poor indicator of population status in this highly clonal species.

Although the LAI did not show any significant trend following the 2005 hurricane season, the slope of the regression line over the ‘post-storm’ interval was positive, which can be attributed to growth of extant colonies since total colony count has not changed. The slope of the LAI increase (+0.0009 m² LAI day⁻¹; ESM, Appendix 2c) indicates approximately 11 years ‘recovery time’ to reach the baseline tissue abundance of 2004 (Fig. 2; ESM, Appendix 2c). Prior to 2005, the most recent major mortality event in the Florida Keys was in the fall of 1998 when a major bleaching episode

coincided with Hurricane Georges (Somerfield et al. 2008; Wilkinson and Souter 2008). The combined effects of these two events in 1998 resulted in losses of *A. palmata* live tissue abundance on upper Florida Keys reefs (Miller et al. 2002b; Baums et al. 2003) that were nearly identical to that observed in the present study. The fact that *A. palmata* did not recover to pre-Georges abundance (Miller et al. 2002b) in the 7-year interval (pers. obs.) between Hurricanes Georges and Dennis, suggests that the trends we observed in this study, though alarming, are not atypical for Florida and many western Atlantic reefs (Gardner et al. 2005). Furthermore, the predicted recovery time of 11 years is longer than the expected return time of a hurricane for the upper Florida Keys of 7 years (Blake et al. 2011), underscoring the need for proactive intervention to enhance population recovery.

Effects of specific conditions

In contrast to *A. palmata* populations in the northwest Caribbean in 2005 (Lundgren and Hillis-Starr 2008; Muller et al. 2008; Quinn and Kojis 2008; Wilkinson and Souter 2008; Crabbe 2009), and to other corals in the Florida Keys in 2005 (Manzello et al. 2007) and 2006 (Wagner et al. 2010), *A. palmata* did not experience any thermal mass-bleaching in this study. The bleaching observed in this study was rarely associated with tissue loss and not considered a factor in the observed decline.

White disease prevalence followed a seasonal pattern with significantly lower prevalence in the spring which then increased into summer and fall, a trend which is consistent with that reported for white pox (Rodriguez-Martinez et al. 2001; Patterson et al. 2002). Peak levels of white disease prevalence observed during this study were roughly 30%. Comparing disease prevalence between studies can be misleading due to its variability on fine spatial and temporal scales. However, we can say that the seasonality and gross appearance of the RM classified as white disease in this study were consistent with those observed in other *A. palmata* studies (Rodriguez-Martinez et al. 2001; Schelten et al. 2006; Muller 2007).

White disease is presently considered to be an un-manageable (NMFS 2006) source of mortality based in part on poorly characterized etiology. The majority of the ‘white disease’ observed in this study was not consistent with the published descriptions of the two named diseases, WBD (Gladfelter 1982; Lentz et al. 2011) and WPx (Patterson et al. 2002; Raymundo et al. 2008), known to affect *A. palmata* and was therefore classified as ‘rapid tissue loss’ (RTL; Fig. 5). It is plausible that these gross signs (described by pattern and rate of tissue mortality) represent multiple etiologies or, conversely, that the full spectrum of disease signs observed in this study originates from a single, general or opportunistic etiology (Lesser et al. 2007). Regardless, the

pursuit to characterize etiology must include more exhaustive descriptions of the range of gross signs and progression of coral diseases, address the dire need for consistent nomenclature (Rogers 2010) and provide tractable, definitive field diagnostic tools (Woodley et al. 2003) in order to improve the understanding of disease dynamics in this species and, hopefully, point to potential control or mitigation measures.

A substantial amount of tissue loss was consistently attributed to snail predation. Snail prevalence in our study tended to be higher than that reported in other Caribbean sites where the prevalence has been reported to be less than 20% (Bruckner et al. 1997; Schelten et al. 2006; Kramer et al. 2009; Schärer et al. 2009) in the past. However, based on this study and others (Knowlton et al. 1990; Miller et al. 2002b), *C. abbreviata* predation is likely to be inversely density dependant, with prevalence (Fig. 3e) and snail density (based on live tissue abundance; Fig. 4) increasing as the focal population declines.

Fewer than two snails per colony were associated with 27% of the background live area loss. Snails have been shown to vector *Acropora* spp. disease conditions (Williams and Miller 2005) and attract butterfly fish (Brawley and Adey 1982), so it is possible that their impact extends beyond direct feeding. Differences in performance (in terms of disease occurrence and LAI loss or recovery) of individual colonies or study plots could not be definitively linked with snail occupation in the current study. This may be the result of insufficient range in infestation among study plots; all study plots included snails. Likely, the confounding of multiple conditions observed on most colonies and all plots compromises inferential power related to each individual condition and their potential interactions.

Damselfish (*S. planifrons*) are known to remove live tissue in association with their algal lawns, chimneys and nests (Bruckner 2002a) and likely contributed to the decline in LAI observed in our study. However, due to the slow and gradual progression (i.e., not observable as RM), our surveys could not quantitatively resolve the amount of tissue loss nor could we discern quantitative linkages between occupation and overall performance of a colony. Although these fish are directly responsible for the removal of tissue, it is possible that their damage may be offset if, for example, their territoriality discourages corallivorous fishes such as *Sparisoma viride* and *Chaetodon capistratus* (Gochfeld 2010).

Evaluating relative effects

To evaluate the impact of a particular condition on a population, a measure of the frequency (both among individuals and over time) and severity needs to be considered. Repeat monitoring studies often assess the prevalence of the various conditions affecting the corals, but this alone is a poor

indicator of their impact. Single snapshot surveys of condition prevalence may lead to even more misleading conclusions as to the ‘importance’ of various conditions, especially for temporally variable conditions such as disease. Attributing tissue loss to multiple sources at the colony level has not, to our knowledge, been attempted in other coral field studies, thus precluding comparisons of our findings. Although these calculations are inexact, they represent a diagnostic monitoring approach (sensu Jameson et al. 2001) that allows useful resolution of the actual causes of population decline and can serve to guide the formulation or evaluation of conservation or remediation plans.

Although snail feeding was the condition most commonly associated with RM in our study (Fig. 5), it is clear, based on calculated tissue loss, that they had less impact on the local population than white disease and breakage. This finding could indicate that they are not promising candidates for conservation efforts. However, snail feeding was remarkably constant over time with respect to both prevalence and amount of ascribed tissue loss. The absolute amount of tissue loss specifically attributed to these predators at each survey has remained consistent at roughly $0.7 \pm 0.26 \text{ m}^2 \text{ LAI}$ (mean $\pm$ SD for all tagged colonies) in spite of a dramatic decline in the number of tagged colonies and LAI (Fig. 2), suggesting that their feeding is chronic and somewhat independent of the *A. palmata* abundance.

White disease and breakage were consistently present and could be considered chronic sources of mortality; however, both experienced acute episodes of elevated impact, which often co-occurred (e.g., winter 2004, summer 2005, and fall 2008). To put catastrophic events like the 2005 hurricane season into perspective, it is appropriate to compare their impact with the chronic effects of disease and predation. For example, the live area lost to skeletal breakage during the few months of the 2005 hurricane season ($29 \text{ m}^2 \text{ LAI}$) was nearly equal to the total lost to white disease ($30 \text{ m}^2 \text{ LAI}$, including both chronic and elevated/acute events) and double that lost to snail feeding ($14 \text{ m}^2 \text{ LAI}$) over 7 years. The tissue loss from the acute white disease event associated with Hurricane Dennis (summer 2005, Fig. 7) claimed an equal amount of tissue ($14 \text{ m}^2 \text{ LAI}$) as snail feeding claimed over the duration of the study ($14 \text{ m}^2 \text{ LAI}$).

Implications for recovery and management

Over 7 years, the three most damaging conditions have resulted in the loss of almost half of the 2004 baseline abundance of *A. palmata* across our study sites in the upper Florida Keys. High rates of mortality in this population persist in both chronic and periodic catastrophic modes. Meanwhile, population growth rates are low, both in terms of new colony recruitment (Williams et al. 2008) and in terms of expanding tissue abundance (Fig. 2). Hence, the observed

recovery rate of 11 years is potentially greater than the return time of catastrophic mortality events in this region, and possibly most areas of the Caribbean where 1998 and 2005 represented acute mortality events from mass bleaching across broad areas (Wilkinson and Souter 2008). Population recovery to pre-1980s levels is unlikely under these conditions; hence, proactive interventions may be required to alter the overall (albeit pulsed) negative population trajectory and maintain/recover ecologically functional populations. While the interventions to enhance recruitment (e.g., propagation of fragments in field nurseries for restocking; Lirman et al. 2010) are being considered for *A. palmata*, this is a relatively complicated endeavor. Interventions to limit mortality would certainly compliment outplanting efforts, but might also be more straightforward and have more rapid benefits if applied to the standing population.

Although storm-associated breakage cannot be prevented, there are low-cost and non-invasive means to mitigate the impact by salvaging and stabilizing fragments with poor probability of self-attachment (Williams and Miller 2010). Although some of the fragments lost to breakage will naturally remain in the study plot (or within the broader population at the reef scale), we found that fewer than 5% of observed fragments recruited to the study population between fall 2004 and spring 2007 (Williams et al. 2008), suggesting that a pool of fragments are available for rescue. Snail feeding can be prevented by removing snails from *A. palmata* where a relatively small number (<2 snails per colony, on average) are responsible for a substantial loss of live *A. palmata* tissue. Miller (2001) estimated tissue loss to snails at $3.37 \text{ cm}^2 \text{ day}^{-1}$ per occupied colony and found that removal of snails decreased tissue loss to the point that it was not significantly different from unoccupied corals. Compared with other corallivores, *C. abbreviata* are relatively sedentary and long lived (Johnston and Miller 2007), suggesting that their removal could provide a tractable management action. Based on the calculated impact of these three conditions over the course of this study, proactive elimination of tissue loss to snail predation and recouping approximately half of the breakage tissue area losses would have balanced the impact from disease. While this may seem to be an unattainable target for managers, in areas where the population continues to contract, such local scale proactive measures become both more feasible and more crucial to the conservation of this iconic species.

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